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H-SCOOP

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BITS & PIECES

* No more 10 page newsletters! Talk about bummers. You may have noticed your envelopes last month looked a bit strange with postage on opposite corners. I had 3 large boxes delivered to the post office full of newsletters for last months mailing. They kindly called me to inform me that they were over the 1 ounce limit for first class mail. I went down to check, and sure enough, each newsletter was just over the 1 ounce hairline, about 1/40 of an ounce. They would not let it pass and I had to take them all back and put an additional 17 cents postage on each one! Nightmare in the daytime! I guess they were out to get me since I wrote a very negative editorial in our local paper a few years back about the poor postal service. You can't fight or criticize the government operations.

Strange thing is, I have mailed 10 page issues many times before without problems. I guess with my relatively new "double density format" there is so much ink on each page, it actually adds up to noticeable weight. In the future there will be 8 page issues, and 16 page issues to catch up. A very hard way to learn.

* Ever wonder how many words are in an issue of this newsletter? Last months issue was over 66,000 words.

* I purchased a very limited number of dual drive brand new **H17 drive cabinets** complete with 1 5" drive (the other side has a blank panel and can accept a drive later). For those interested, I will sell the cabinets alone for $170, cabinet with one drive for $300, or new 5" 40 track Wangco drive for $140. These are very limited, call if interested.

* The **hard drive project** has been held for evaluation. It was just called to my attention by Dave Troendle that a 5" 40 Meg hard disk drive has just hit the market, and at a good price. If the technology continues, and I believe it will, the price of hard disks will drop substantially in the next years time. I also found out that a few other larger companies are doing a hard drive, so we'll wait a month or so and see what happens.

* The cat's out of the bag. I mentioned last month that I had reason to believe we may have another **major magazine** devoted to the Heath/Zenith systems. I could not say more than, but I received it in print now. Most of you have probably received the mass mailing from Wayne Green for making money converting TRS-80 software to Heath/Zenith format for a nice royalty fee. I hope this changes things, but thus far Instant Software for the Heath computers has been a real joke. I know of several individuals who have written in expressing their willingness to participate in this venture, with no follow up results. I would be interested in other individuals comments along this line. Anyway, further in the blurp Wayne said "If you send in enough Heath/Zenith articles and programs I'll spin off a Heath magazine." And I wouldn't put it past Wayne. For those of you who do not know, he is publisher of Microcomputing, among many others. If there is a buck in it, count on Wayne to be there.

* Feedback on last month's BIT for speeding up the Siemens drives. Quite negative. Several wrote in to say it didn't do anything, some wrote in to say they realized only a few ms increase in step time, but it appeared the steppers were really being driven hard, causing a lot of noise. Oh well, you can't win them all.

* Charles Loeb writes in to tell us that the **CUBIS PS ASCII printwheel** for the Diablo 630 has a Diablo part number 311901-01, and is available from Visible Computer Supply (800) 323-0628 for $75.55. It is also available as a Xerox Part (Diablo's parent company) 9R21801 from the Xerox Regional Distribution Centers for $50.

* **Extended Technology Systems/** 811 Warren St/ Reading, PA 19601, is offering a service for alignments on 80 track Tandons. Tentative charges are $45 if drive can be aligned to specs, and $25 if they can't be, with a note of parts needed for repair. They will not repair them. Contact them for prior arrangements.

They are also extending their offer till October 1st for the products mentioned in the May issue of H-SCOOP. These include the **ETS-C2.2 ZCPR** Z80 CCP for the CP/M operating system, **ETS-H2.1** which is a new SYSCMD.SYS for HDOS 2.0, and the **ETS-64** which is the H89 16K RAM expansion set.

* Bob Todd wants me to let you know that he will be at the HUG convention in August with his **SIG/M** and **CP/MUG library** on 5" disks. Anyone interested in having part or all of the library downloaded, bring the diskettes along, either in 5" 80TK DS SD Livingston format, or 5" 40 TK SS SD Heath format. Be sure to have the diskettes all ready formatted to save time, and bring enough along. For more info on this, refer to March and April issues of H-SCOOP.

* More on the **Percom DD H89 controller.** Bob Sternowski/ 2550 Larick Dr/ Marion, IA 52302 has a hardware mod to allow Heath drivers to be used with their parallel port for either HDOS or CP/M. If interested send a SASE. He further comments that the auto disk type recognizer feature is great. You can insert a hard or soft sector disk in the drive and the system will automatically know which one has been inserted and adjust itself accordingly. Bob and others have commented on how slow the board is. He did a test of PIPing a file under HDOS from hard sector to hard sector -- 2 seconds read and 4 write. Soft sector to soft sector took 4 seconds to read and 11 to write. Under CP/M, hard to hard took 3 seconds to read and 8 to write, while soft to soft took 7 seconds to read and 25 to write! This is indeed strange, since double density soft sector transfer rate is double the single density speed! In fact all my double density controllers are much faster in any mode for transfer than the hard sector controllers.

He also was annoyed at the coarse drive step rate choices of 6, 12, 20, and 30 ms only. The large HDOS SY device driver is also very large at 26 sectors.

A correction to last months feedback on the Percom drive controller. The H17 board is needed for hard sector disks, and the Percom board is needed for the only soft sector. It can't do both.

* Terry Smedley called to inform me that the Heath utility **ONECOPY**, as supplied on the HDOS distribution discs WILL work with the 80 track tandons, 2 sides, if this drive is set up as the BOOT drive.

* Those who have ordered the **CP/M utilities** from Quikdata by **Elektrokonsult**, be patient. They should

have gone out by the time you are reading this. There were major delays in mail and more in bureaucratic red tape customs. They were not delivered here, but were held up in Milwaukee customs by the Treasury department, which is about a 120 mile round trip for me to pick them up, not to mention paperwork involved. Our government tries to make it as difficult as possible to import from foreign countries, and charges plenty of duty tax to claim items, not to mention the cost of an agent for filing customs declarations - another real nightmare!

* A note to those installing Heath's 2.2.03 CP/M. All 3 ROM's are needed in the H89. Previously you only needed one, the others were mainly for the H47. If you just installed the one and can't figure why the CP/M won't work, that's the answer.

* Note from Extended Technologies. Before having any drive aligned, be certain you copy all files from all disks made on that drive to another drive you have which is readable. Chances are, after alignment, your diskettes made up in that drive will no longer mount or be readable.

* Bill Abbott called to tell me about the **H89 expansion interface box by Microflash** as written up in the March issue of this newsletter. It's about 6" high, 12" wide and 16" deep, made from heavy steel with an excellent layout. It has a quiet fan, and makes additional buss space available for the H89 computer, with 100 port decode lines. Bill said he was very happy with its design, looks, and operation. For those who missed the March issue, you can get this unit from Microflash Co/ 4916 B. Carol St/ Skokie, IL 60077/ (312) 677-4928.

* **Evryware**/ POB 60802/ Sunnyvale, CA 94088/ (415) 321-2708 wants to inform all that their software is now available for CP/M on soft sector disks. Refer to REPORT last month for full story on their software. Available from them or most Heathkit stores.

* Terje Bolstad of Elektrokonsult reports that his CP/M disk utility **DTEST** is a good general purpose utility to use to find problems on diskettes and on drives themselves. When using it to test a drive, large number of test passes of both read and write should be specified. If the same errors occur in the same locations, you are bound to have a drive problem.

* There seems to be some **compatibility** problems **between DG and Trionyx** equipment. One individual called who has the DG memory and Trionyx Z80 CPU. He said a small mod was required to the DG memory to make it work with the Trionyx Z80. Call DG for more on this one, I understand they have the fix. I was also told by several individuals that the DG memory board foil is very delicate. When attempting to jumper things on the board and/or resolder, the foil and solder pads have been lifting off the board. If you are going to work with this board, be very careful. A few individuals also called me to tell me that when asking DG for advice, when DG was told the system contained DG and Trionyx equipment, they were quite irate and huffy about the whole situation, as if to say "what are you doing with Trionyx equipment in the same machine with DG equipment".

Another individual felt Trionyx was wrong in using port 077Q for their CPU board as DG had been using it first in their system. Thus again, if you have the Trionyx Z80 board, and the DG board which uses the same port, there is going to be a conflict. DG feels Trionyx may have done this just to spite them. Whatever, there seems to be a war going on between the two, and perhaps they should both grow up and get together and be compatible, sit down and talk things out and keep their systems compatible yet not conflicting. The attitude expressed thus far has been very unprofessional. The shame of the matter is both companies have some very excellent products out.

* Chris Simmons who gave information in issue #26 on the **UCSD Pascal SPIN** program sends in a revision. He said he gave me a bad version to print in issue #26. Here is the corrected version.

```
.PROC   SPIN
MVI     A,15H    ;disk spin time
STA     2048H    ;write to memory
XRA     A
RET
 .END
```

Any further questions on this, address to Chris at 2802 N Greenwood St/ La Grande, OR 97850.

* Feedback on the **Disk drive service centers.** First let me say that anyone who purchases a drive through me can get it serviced through me, and usually at little or no cost. Individuals must not be understanding my policy, because they are having bad experiences with these so called disk drive repair centers, when they could have sent them here. I was told by Patrick Lindsey to stay away from **Computer Service Center**/ 1023 N. LaBrea/ Hollywood, CA 90038 as written up in the March issue. They advertise a 3 day turn around time. Pat sent a Tandon in and it took over a month, cost him $88, and when he got it back, it still did not work. They also did not have the brains to ship it back with the head protector installed. **Stay away from them!** Another one that comes in with bad recommendations from the March issue, is **Micro Computer Repair Center**/ 9 Main St/ Danbury CT 06810. George Duke sent in a Wangco and was quoted $29 plus parts. They advertise 72 hour turn around time. After 10 days, he called and was told it would go out in a day or two. Another call back a week later, he was told there was problems with parts and it would go out at the end of the week. Next time he called he was told they raised their base price to $45 but would honor the $29 for his drive. After still more calls and over a month later he got the drive back with a $45 bill. He said the drive did work so he did not pursue it any more. Just a warning, watch out for these places. If anyone has found a reputable place, please let us know.

* Individuals with the old Tandon drives without the door start switch have been asking about modifications to bring it up to the door switch models. I have contacted Tandon and was told it would involve quite a bit and be costly, so you're better off to forget it.

* After waiting over 2 months for my part from Heath for my H89, I finally received it. All I have to say, is a $6000 business system down for 2 months for a $10 part is not at all very impressive service or support.

* We are in the process of evaluating some good, easy to use, low cost CP/M **business software.** Included in this is a MBASIC CP/M small business **payroll** system. It's very complete, menu operated, and will handle up to 32 employees. Does W-2's, taxes, etc. Another is the full Osborne **integrated G/L, A/R, A/P** package written in CBASIC. This version has been fully debugged, enhanced, re worked for the H19/H89 screen, and IT WORKS. Look for a $150 price for payroll and a price of around $195 for the integrated package. More information when all testing and evaluation is done.

* **Magnolia Microsystems** now has the **new CP/M** out. This version makes the Magnolia card compatible with the H37, H47, and H67 formats, supporting over 28 disk

formats, including the original Magnolia formats. It also supports the H37 card. Since the H47 has been dropped, Zenith is recommending the Magnolia systems to their clients. This new version also supports type ahead buffer. Many utilities have been improved and revised. All Heath stores have some Magnolia products, either for their own use, or for sale. They will all receive an update disk, as will all dealers, which will enable anyone to update your disk for you. Present policy is free update for all systems ordered within the last 4 months or so. This includes diskette update and manual update. There is a $25 charge for a PROM which is only required if you want to boot this off a Z90 5" soft sector system, otherwise the PROM is not needed. Any version of MMS CP/M that has been shipped to dealers and sold before February 1982 will carry a $25 update charge for disk update, disks required for exchange, and $5 for new manual.

* The following **Product Status Announcement** sent in by Dean Gibson of **UltiMeth** Corporation.

"The March 1982 issue or REMark magazine from the Heath User's Group contained source modifications to the DKH17.DVD device driver (which HUG calls "SY.DVD") which we wrote and then DONATED to HUG. Two of the modifications are based on misconceptions about our code, and one of those modifications can cause serious problems when subsequent diskettes are INITialized and then SYSGENed. Since HUG made these changes without contacting us and has assumed full responsibility for the maintenance and user support of DKH17.DVD (it was this stated requirement of HUG to be able to modify user submitted software that led Livingston Logic Labs to withdraw its submission of BIOS-80 (a product like DKH17.DVD for CP/M) to HUG), we are adapting the following policy regarding HUG'S "SY.DVD":

1) Effective July 1, 1982 we will no longer support the HUG "SY.DVD", modified or not. After that date, all questions regarding that product must be referred to HUG.
2) We will continue until August 31, 1982, our policy of allowing users who have the HUG "SY.DVD" to return their original distribution diskette and $20 (CA residents add sales tax) to us for an upgrade to DKH17V2.DVD. After that date, upgrades will only be available to those who can show (via receipt) that they have bought HUG's "SY.DVD" within the prior 30 days. DKH17V2.DVD has been available since October 1981, and contains the enhancements by HUG (properly done) and others, including 4 MHz support and unloading the disk heads when RESETting a diskette. Source code is NOT included."--Dean Gibson, President UltiMeth Corporation.

* A comparison of the 8086 16 bit CPU with the Z80 CPU by Dean Gibson. "Last summer we had a consulting contract to implement CP/M-86 on an 8086 running at 5 MHz, with a 16-bit bath to 704K of RAM memory. When Digital Research provided CP/M-86, they provided two assemblers for 8086 source code: one assembler runs on the 8086 under CP/M-86, the other assembler runs on an 8080 under standard CP/M. Using 8-inch floppy diskettes on a modified H89 running at 4.096 MHz, the assembly of the BIOS for CP/M-86 took 3 minutes flat. Running at 2.048 MHz, the assembler took 5 minutes, 45 seconds. Using the same 8-inch diskettes on the 8086 running at 5MHz, the assembly of the same BIOS source file took 3 minutes, 45 seconds. In all tests run, disk I/O time was less than 10% of the total run time. Prospective purchasers of 8088 systems should note that the 8088's 8-bit p?th to memory would further slow any comparisons.

* J. Norman Goode, publisher of **Micro Moonlighter Newsletter**/ 2115 Bernard Ave/ Nashville, TN 37212, announces a **summer Writer's Contest** which will award a personal computer to the winning entry. In keeping with the theme of Micro Moonlighter, we are looking for articles which describe the best use of a personal computer in a home based 'cottage industry'. The business opportunity of the 80's is a cottage business with a personal computer as its center. There are many individuals and couples who have started a successful enterprise using the latest in technology, and we want to make our readers aware of these pioneers.

The winning entrant will receive a choice of a personal computer and/or software valued to $500. The winning entry will become the property of Micro Moonlighter and will be published within its pages. All entries will be reviewed for possible future publication, where articles selected will be purchased at three to five cents per word, depending on length, content, etc. Deadline is October 31, 1982, with winners announced on November 22, 1982. Acknowledgement of receipt of material will be made if entrant submits stamped self addressed postcard with entry.

NEW STUFF

SOFTWARE RELATED

* From **Gregory Marsh**/ 5279 Miles Ct/ Woodbridge, VA 22193/ (703) 590-3360 comes software and doc files on disk to **operate up to 6 5" disk drives** with the H8 and H89 single density Heath controllers. Includes instructions for single chip mod to the H17 type disk controller. Will work with: 1) Standard HDOS 2.0 SY device driver; 2) Original "HUG"/Ultimeth SY 400K driver; 3) UltiMeth DKH17V2.DVD; 4) Standard Heath CP/M 2.2.03 BIOS; 5) Livingston Logic Labs BIOS-80V2 (Same as BIOS-80 2.2.03). This software is fully compatible with Standard HDOS or CP/M commands and no changes are required to anything other than device driver or BIOS. Same package as Ver 3.0 of FOUR DRIVES FOR HDOS, with CP/M added. Cost is $20, updates $12.

Greg also has available a **new SYSCMD** for HDOS, **GSYSCMD** for $30 (previous purchasers of NEWSTAT may deduct $5). GSYSCMD, which replaces the HDOS SYSCMD.SYS, puts an end to the need for typing the device name when running programs. Allows you to run programs on any SY: device without the need for specifying the drive the file is on. Supports the file search function for up to 6 drives. The Environmental Control Systems PC-12 and PC-89 real time clock is also supported by a new TIME command. The popular NEWSTAT program from Marsh is also included to replace the Heath STATUS command. All MOUNTS, DISMOUNTS, and RESETS can be done using the H19's special function keys. Same size as the Heath version. Includes source code. Does not modify HDOS in any way and can be easily removed any time.

* Software from Marvin Youngstrand/ POB 15882/ Honolulu, HI 96815/ (808) 523-8971. **CLOCK2.BAS** is a functional program displaying full screen large characters of time and date when your system is not being used for other things. Computer rings bell on half hour and hour. $29.95 plus 10% shipping & handling. Requires HDOS, MBASIC and 48K. Supplied on 5" hard sector.

SIM2.BAS is a program to develop a set of tableaus based on linear programming, simplex method. Each screen shows the development of each independent tableau, including pivot points and new line development, with last tableau displaying the complete solution. Can be used to solve complex problems using constraints. $29.95 plus 10% shipping & handling.

Requires 56K, MBASIC and HDOS or CP/M, whatever you specify with order.

* Hank Labarbara/ 3803 Isabell/ Skokie, IL 60076/ (312) 675-5223 has developed a program similar to TEST17 which works on the 8" Livingston controller for the H8 and H89. Includes general drive checkout, bad sector media check, and hardcopy error report. Shipped on 5" hard sector unless specified otherwise. $24.95, which includes shipping.

* **Lindley Systems/** 21 Hancock St/ Bedford, MA 01730/ (617) 275-6821 has available three new printer **drivers** for the **NEC PC-8023-A printer.** The first is an HDOS driver with standard driver features plus more. The second is a CP/M version of this driver. Each of these are $22 postpaid. The third is an HDOS UPC (user programmable character) driver for $25 postpaid. This one has the H19 character set programmed in. The user can program any other characters into it. Robert Lindley says there is a money back guarantee on the drivers.

* **Generic Software/** POB 1154/ Troy, MI 48099/ (313) 879-6903 has quite a list of software. Write if you want their catalog. They have a very excellent graphics demo disk out for $6 (5" CP/M) which demonstrates their complete product line. They just came out with a new product, **LOAN MASTER.** This is a loan amortization type program to calculate payment, balance, interests on loans and mortgages. I mentioned two others in the February issue. Loan Master is different in that is uses a unique "forms" mode for data entry. It can solve for any unknown loan parameter. Capable of providing either annual or periodic amortization schedule. Data can be outputted to printer, console, or disk file. Requires H8/H19 or H89 with 48K RAM and 1 drive, either HDOS or CP/M (specify which, LOAN MASTER-H for HDOS and -C for CP/M). Available on 5" hard sector for $29.95 from Generic Software, or at most Heath stores.

I ran this one and it produces neat hard copy output in 8 columns; period, date, payment, interest, interest to date, equity, equity to date, and balance, with all summary info printed below.

* From **Apogee Software/** Box 15124/ Savannah, GA 31406/ (912) 925-3765 comes a new arcade game, **FROGS.** Guide your frogs through rush hour traffic, from log to log across a swiftly flowing river, and into their havens. Don't tire them out with too many quick jumps, but avoid the alligators, snakes, diving turtles, and quicksand. The longer you last, the harder it gets. $14.95 plus $1.50 shipping, distributed on 5" hard sector or any H37 format for an additional $2. Specify HDOS or CP/M. We will be doing a review of this in a future issue.

* **NEWLINE SOFTWARE/** POB 402/ Littleton, MA 01460/ (617) 486-8535 (eves) just published a new software catalog. Contact them for a copy if you did not receive one. They have added a new program, **H25 GRAPHICS GENERATOR.** This is a sophisticated interactive graphics screen editor for creating and modifying graphics drawings on the H19 terminal. In addition, the graphic displays can be sent from the H19 screen to the H25 printer. Utility included to print continuous pictures from disk. Use this program to design your own invoices, statements, letterheads, or just for fun graphics. Written in 8080 code for speed. Includes features such as cut and paste, put to or get from disk, fill area with any character or symbol, insert, delete, cursor motion, and more. Requires 32K memory, $44.95

* DB SYSTEMS, Inc/ 79 Marshall St/ Providence, RI 02909/ (401) 331-5034 has a new software package, **SY67.DVD.** It contains 4 HDOS device drivers for the Z-67 winchester disk drive system. They also have two Unix like utility disks for use with the H47 or H67, and a MODEM communications package. No price given, contact them for more info.

* **ACCESS** is a software **modem** program for the H89 computer by Hilgraeve Inc/ POB 941/ Monroe, MI 48161/ (313) 243-0576. This is a CP/M MODEM program requiring a serial port, 0-org CP/M and 48K, 5" hard or soft sector disc. Price is $29.95, with "C" source code available for $5 extra. Manual alone is $5. Features include the following: Disk storage if incoming text can be toggled on and off with a single key; All parameters and options can be manipulated from within the program; Single key toggle to echo display to a printer; special function keys used; ability to remove LF codes at EOL when transmitting, or add LF to CR's when saving file; much more.

* **HOYLE & HOYLE** SOFTWARE/ 716 S Elam Ave/ Greensboro, NC 27403/ (919) 378-1050 advises that their color graphics games **Brick Break** and **Rapid Recall** ($16.95 each) now run on the **Entertainer,** the H89 color graphics package by Sandia. They will also run on the HA-89-3 board by NOGDS when it is released. **Steer Kleer** ($14) is now available in CP/M format. Maps, hints and scoring guides for **A Galactic Experience** and **A Remarkable Experience** sell for $18.95 each.

A new adventure available from Hoyle is **A PHYSICAL EXPERIENCE** for $19.95. Requires H8/H19/H17 or H89, HDOS and 32K. This is a very challenging adventure for experienced explorers. Don't let the radiation get you, while you prevent a supernova on the planet Secundus around the star Betelgeuse.

They are running another contest. $100, $50 and $25 will be awarded to the first three people who complete the game and send in the final message. The winners must reside in different states, and answers must be submitted on entry blanks which come with every game. Earliest postmarks win. No hints will be given out.

* From **MICROTRAN Co./** 76 Flintwell Way/ San Jose, CA 95138/ (408) 226-4122 comes various programs for users of HDOS Microsoft FORTRAN. Some of these will be of interest to other language users, since the FORTRAN file is compiled in an .ABS code.

MICRO-CORE is a FORTRAN callable graphics package of the future with many state of the art capabilities including: device independent; can be interfaced to four different graphics devices simultaneously via device drivers; works in user, not device, coordinates for your convenience; uses the "view window" concept to perform auto scaling, and supports auto window clipping; and is easy to use. Price is $62 which includes shipping and handling, and includes source code.

MICRO-CORE driver for Percom Electric Crayon is a device driver for use with Micro-Core to interface a Percom Electric Crayon to the H8-2 parallel board. Price is $30.50 which includes shipping. Other drivers are being developed. If anyone is interested in loaning MICROTRAN your other hardware graphics, they will develop a driver for it. In turn, you will get a free copy of the driver plus a free copy of MICRO-CORE. Call if you are interested.

ADSORT is a FORTRAN callable sorting library which is VERY fast and can sort all types of data. Uses a previously untapped sorting technique that is commercially unavailable anywhere else on any system. Cost is $30 which includes shipping. Complete doc and source code allows implementation of this fantastic sorting technique in other languages.

CLIMB is a free format Console Input Library. Permits integer, real, text and HDOS filename data to be input in "user friendly" free format with no fear of I/O format errors. Makes writing interactive programs much easier and results in superior user interface. Price is $22 which includes shipping. CA residents add appropriate sales tax.

HARDWARE RELATED

* Update from **Ray Albrektson,** address in classified. He is now selling assembled and tested **H89 clock/ sound** boards for $90, or bare boards for $22 (plus $5 for HDOS software). Assembled board including software, but minus MSM-5832 clock chip, crystal and sound chip for $55.

* **PHILLIPS ENGINEERING** CO/ 721 Pleasant St/ St. Joseph, MI 49085/ (616) 983-3935 has available a **Digital Voltmeter Board** (DVM) for the H89 computers. This 8 channel DVM board is capable of professional laboratory data acquisition and control functions, yet at its low price is well suited to home and personal applications. Plugs into right hand slot of H89. Accurate and stable over long term measurements. Easily programmed in MBASIC or BASIC for complete I/O control of numerous data points and instruments. The data can be stored, used in computer analysis, printed out, etc. Features include 8 single ended inputs, software selected; 12 bit resolution with polarity (nominal +/- 4V in 1mv increments); 7.5 conversions per second; auto zero; 1-10 V variable voltage range. Add on technical assistance includes temperature probes, pressure plug in module, pH plug in module, flow rate plug in module, and support for Home, Business and plant security. Price is $189.50.

* Don Long of **Environmental Control Systems/** 9319 Willowview Ln/ Houston, TX 77080/ (713) 464-1717 is making available **IR detectors** for use with a parallel port. They are a major breakthrough for home control and security applications. Both units have self contained Lithium batteries for 5 to 15 years continuous operation, and have two leads as the output, no input required. The output lead is a relay closure which activates when there is detection of heat and motion (people detector) in the set area. Passive IR sensor is superior stability and performance made possible by design. Intruder must first enter one viewing area and then the other to obtain an alarm state. Any simultaneous activation of both channels caused by false alarm conditions will be cancelled out. Typical battery cost when replacement is necessary is $2. Because power is self contained, much time and hassle is saved when wiring unit since high voltage AC, or power supply voltages are not needed. You simply bring the two signal wires to the parallel port. Can be set to be Normally Open or Normally Closed. Can be flush mounted such as in a standard single outlet box or hole in wall, surface mounted or corner mounted. Superior rejection to White Light, immune to RF, line transient and lightning, and contains automatic battery fail-safe system.

Two models available. The R50 has a range to 50 feet with a 5' protection area at this distance. It sells for $66 single quantity. The RW-25 has a 25' range, but covers 75 degrees pattern spread, or over 30 feet. This is $78 single quantity. Both units will switch up to 100ma at 24VDC, ideal for parallel port operation, or directly controlling small bell type devices. 3 year limited warranty. Order from Don Long or Quikdata. Allow up to 3 weeks for delivery, depending on stock.

Don wants to mention the latest HDOS release for the PC-12 and PC-89 clock parallel boards is release 3.5. This support now allows easy reading of and writing to the clock as a device driver directly from the HDOS command mode, or from programs. The driver will now also constantly display the time on the H8 display, or H89 screen. Also included is a diagnostic PPI test for testing the interface components of the boards. The CP/M distribution disk includes a new menu driven .COM file to set and read the clock. Either disk is available for $10 media and handling charge.

* A new **serial/ parallel I/O board** is available for the H89 from J. Hassall of H & H Enterprises/ PO Drawer H/ Blacksburg, VA 24060/ (703) 552-0599. The board supports the standard LP: and AT: ports of the H88-3, and adds 2 serial ports- three parallel ports which are user programmable for any combination of the 24 I/O lines (3 ports of 8 bits each). Parallel ports use the 8255 PPI.

Bare board, 1222 is available for $48.50. KC1222 for $128 has all necessary items for complete serial/ parallel I/O board. KP1222 for $97 has all necessary parts except serial chips and cables from the H88-3. All prices include doc and are shipped postpaid. Write for quantity discounts, VA residents add appropriate sales tax.

They also have device drivers available for the board for parallel printers such as MX-80, Centronics 739-1 and 739-3, and NEC PC-8023A is coming

* This section will cover some product info and discussion on double density floppy disk controllers.

From **MI-8/** 822 E County Rd 30/ Ft. Collins, CO 80525/ (303) 669-4116 is coming out with **THE DOUBLER.** This is a plain and simple 5" DD controller for the H8 and H89 systems. It will read and write 40 or 80 TK single or double sided drives. It will be available after mid September and sell for $249.95. Orders are now being accepted at a 25% reduced rate of $187.47 plus 5% shipping and insurance. Orders will be shipped on a first received, first shipped basis. Visa, MC, check accepted. Write and ask for 1982 catalog.

CDR SYSTEMS INC/ 7667 Vickers St, Suite C/ San Diego, CA 92111/ (714) 275-1272 is coming out with a DD board for the H8 computer. They were scheduled for July, but put it off till September to allow for more testing before release. This one will sell for $495 and is a do it all card. The way it looks, this is the one I recommend for those who want both 5 and 8" drive handling capability. In fact, I may pick this one up myself for my systems, and to distribute. It will work with the standard or Z80 CPU at 2 or 4 MHz, using DMA for reading and writing to memory. It will handle all 5 and 8" drives including the 80 track 5" drives. Although it is a soft sector controller, you can use hard sector 5" disks, it simply ignores the extra holes. CP/M will be supplied, as will a new BOOT ROM. HDOS will be out shortly from Livingston Logic Labs.

The BIOS is currently set for up to 4 soft sector drives through this controller (**FDC-H8**) and 3 hard sector drives through the standard Heath H17 controller. Both controllers can be connected to the same set of 5" drives. The hardware can handle four 5" drives, AND four 8" drives. It is compatible with the H17 controller system, and has the capabilities of both the H37 and H47 combined. Of course the standard CP/M format is supported. So with Magnolia's new CP/M, it looks like finally we will all be compatible!

CDR's H89 card has been very successful. I have heard of no complaints against them, on the contrary, much praise for their support and service. They also seem to know what they are doing, and are doing it right. That is why I feel they have the best thing going for the H8, and why I may be willing to pick them up as a dealer.

Whatever happened to Mako? I don't know. I, and everybody else, just stopped hearing from and about them. I guess they ran into problems, and we'll hear from them again.

What about the Trionyx H8 board, the **C-H8**? Prices are $425 assembled and $350 kit, projected shipment around end of September. Not to be negative, but I doubt if they'll make it. They also claim 8 floppys at a time of both 5" and 8" variety. They also say it will handle SD and double density, and 40 or 80 track 5" drives. They also say it will have Winchester hard disk drive support on it, which will be nice, if all that will fit on one board. DMA will also be used here for data transfer, rather than ports, same concept as with CDR. What I don't like about their add is "Using DMA, disk operation will proceed at much higher speeds than currently in use with the H8 computer". Not necessarily true! Disk transfer rate is determined by the disk drive and format. For example, a single density 5" drive transfers at 1 byte/64 ms. A double density 5" drive transfers at 1 byte/32 ms. 8" SD rate is the same as 5" DD, and 8" DD is 1 byte every 16 ms. This is within the transfer port specs, and this is the determining factor.

Getting back to Trionyx, the C-H8 will be configured from the operator's console using self-prompting system software. Configuration is a one time only operation, although changes can be made at any time. There will be no configuration switches or jumpers needed on the board, this sounds good. This board also will be able to be used with the standard 8080 CPU, or the Z80 CPU. They plan to support both HDOS and CP/M. For more info or to order, contact Trionyx Electronics/ POB 5131/ Santa Ana, CA 92704.

* CDR SYSTEMS also NOW has available a **hard disk** for the **H8** and for the **H89**. The 10 Meg unit sells for $3195, while the 5 Meg unit goes for $2495. It now supports CP/M, and will eventually support HDOS as well. Transfer rate is 5 million bits per second. It looks like a very good system.

* Floppy Disk Services also has a hard disk drive system for the H89. The 10 Meg unit is about $2700.

THE "Z" MACHINES

Presented here is the information on the new Heath/ Zenith "Z" machines, along with comments, observations and projections. Part of the deal I made with Heath was to see the machine so I would know the whole 'scoop' on it. In return I would stop printing rumors and any more about the "Z" machines, and Heath would notify me when I could release this info. Well, that never happened. I found out myself after the rest of the world knew about it. I was quite upset that the "deal" to notify me when release was OK was not carried out by Heath as agreed upon, thus no more deals like that.

First of all, there are two Z computers, and there will be an entire product line eventually devoted to the Z100 line. The computers are the most sophisticated computers ever built, being shipped 100 per cent tested with a 48 hour burn in. All factory built units will be FCC certified class A, with kits being certified class B. This should eliminate the RFI problems we have had with H8's and H89's, especially with TV reception. The names are the **Z110** for a low profile model without attached CRT, and the **Z120** being the all in one unit with integrated CRT, otherwise functionally equivalent. The low profile unit is about 7" high, and weighs about 40 pounds. The all in one weighs about 50 pounds. Both occupy about 20 by 20 inches of desk space. Both also can be set for 110/220V and 50/60 cps operation, 400 watts max.

The Z computers contain an 8088 and 8085 processor both operating at 5Mhz, which helps make this a very fast machine. Unfortunately the 8085 supports the 8080 code, but not Z80 code. Most of the 8080 software from the H8 and H89 should run on this system. Z80 will not. I realize Heath never released Z80 software, but some vendors have, and for those having a lot of Z80 software, this may be a problem. The 8088 is the 16 bit chip processor part of the machine, and because of the 16 bit structure and its powerful instruction set, it will be VERY fast, many times faster than the H89 Z80, between 2 and 10 times faster.

In a nutshell, both Z machines will have dual horizontal 48 TPI drives (Tandon TM-100-4's were installed during my visit). This translates to 320K disk storage per drive. Since this is a 16 bit machine, it will allow large memory addressing, far beyond the 64K we are accustomed to on the 8 bit machines. The CPU board is very complex, being a 4 layer PC board. It contains keyboard encoder, two on board serial ports capable of operating at 38K baud (with IBM format or standard format for each), one Centronics type parallel port, bus expansion, DD disk controller for internal drives and 192K main memory with parity checking features, probably populated with 128K when purchased. Since the serial ports can be set for either asynchronous and bisynchronous operation, this system will be capable of communicating with mainframe computer equipment. Total RAM addressing capability is 760K. The last 1/4 meg is reserved for ROM chips using 64K ROMs, and the memory mapped video. Heath has installed a $500,000 tester for the CPU board for more efficient operation at the plant. Although there will be no math chip on board, it will be possible to add one later via S-100 board if desired. Because of the complexity of this and all other boards, they will all come assembled. You would not even want to solder and troubleshoot a 4 layer board.

The layout of the low profile unit has video board on top of CPU board (both horizontal), with dual drives over all of that. The All in one has the dual horizontal drives stacked to the right of the display. That's what I like to see, HORIZONTAL drives!

The **keyboard** unit is a standard typewriter layout, with 95 keys total, including 16 special function keys and an 18 key numeric keypad. It will operate with a FIFO (First In First Out) buffer, so rapid typing or data entry will not cause any keystroke losses. Every key generates an 8 bit code, thus it is not H19 compatible without appropriate software. It contains its own 8741 processor chip, and like the H89, has audible key click. The RESET switch has an LED for power indication.

Bus structure? They learned their lesson on incompatibility. The new machines will have a 5 slot S-100 bus for interface with the S-100 industry standard boards. Since just about everything is built into the CPU board, it is unlikely that the slots will ever be filled up. But as we all know, more neat things come out and we **do** fill up the slots, so I suppose someone will eventually come out with a S-100 expansion setup. If you like a front panel, I see no reason why you couldn't get a front panel S-100 card with ribbon cable to connect to a front panel display! One slot of the S-100 buss will be taken by a do it all 8" disk controller card for up to 4 outboard 8" drives and 4 outboard 5" drives of any variety (soft sector). This card uses the Z207 controller, according to Zenith release info, but then it also said the 48 TPI drives were 80 tracks per side, so I'm not sure about them.

The non-glare green screen 12" **video display** is a memory mapped video with high resolution pixel addressing capable of 640 dots by 500 lines in the interlace mode (50 lines by 80 columns possible as each character cell is 8X16), and 640 dots by 250 lines in standard mode (24 lines by 80 columns with 8X9 character cell, 25 line "status" like H89). Pixel addressing means no alpha numeric mode, all graphics and characters are produced by dots. Since all characters are generated by software, it is possible to have ANY type of font. Although all characters and graphics must be generated by painting every individual dot, this did not seem to slow down the display at all compared with the H89 operating at 9600 baud. The video will contain its own 96K memory on the video board with expansion to 192K. This board is also a 4 layer board. With the large memory available it will support 3 color graphics, with an external color monitor and software, using 3 planes of memory one per primary color, using 640 dots by 500 vertical lines. We saw one connected up to a color monitor and it was excellent. The color will be standard on the Z110 low profile unit, and optional with the all in one Z120 unit. Don't get too excited though, because a very expensive color monitor is required for that resolution, probably close to a thousand bucks. Both units have standard RGB output, with monochrome composite video output also on the low profile unit.

It may be important to note another reason for the machine's fast speed. All past Heath machines drive the video via serial interface. This is slow compared to memory mapped video. The CPU simply "writes" the data to appear on the screen in a specific memory area which belongs to the display, and the display electronics pick it up directly from memory locations.

Comparison to IBM? There are many similarities between the IBM and Z machines as far as looks, software compatibility, etc., but the Z is by far a much better machine. The IBM we saw had a 63 W power supply, which was not even capable of powering 2 drive motors at once. IBM had kludged the controller card and cable to only have 1 motor running at a time. This means longer access time to get to the drive not rotating. The IBM is also an 8088, so we should see a lot of software compatible between the two machines. In fact, one of the operating systems shipped with the unit is MSDOS, which is MicroSoft Disk Operating System, the same one IBM uses for their personal computer. The IBM does NOT come with serial ports so their 5 slots are taken up quite fast. It also only has 64K on the main board, which again means more slots are used up to bring it up to the 192K Heath makes available. The Z machines have switching power supplies for top efficiency, rated at 300 Watts! This unit will be supplied as a sealed box from another vendor for Heath. Since Heath wisely placed more goodies on the main board, the S-100 slots will be more available to the end user. IBM uses 16K per color bank, which means the Heath units will be higher resolution with better color (higher resolution B & W also). Both the Heath and IBM unit use the 6845 CRT controller chip. The actual color of the cabinets of the Z machines will be close to the IBM color.

What about the processor chips? To elaborate on the above mentioned, the 8088 is the 16 bit chip. The Heath machines will run at 5 MHz on both of its bus'. The 8085 will handle the 8080 present software for the H8 and H89, thus we should be able to run most of the software from the "old" machines on this one. When properly utilized a 16 bit machine can be very fast and efficient, but don't expect to see lot of 16 bit software emerge over night. The present 16 bit machines out have very little software available to support their structure. The only main advantage presently is extra memory addressing capabilities. Within a year or so Heath hopes to have lot of the 8 bit software converted to make use of the 16 bit bus structure. The 8085 is twice as fast as the Z80 at 5MHz, and the 8088 at that speed will be approximately equivalent to a Z80 running at 15 MHz.

As a side point for those interested in mysterious things such as interrupts, the machines feature a 16 level interrupt control, with the 8259 chips.

Disk drives? For the external drives a standard Shugart compatible 8" DD disk controller board can be handled by the S-100 bus to allow you to run double density on external drives as mentioned before. The internal drives are DD drives also, with up to 800K per drive (using DS DT drives). The controller on the CPU board uses the 1797 chip for the controller, which is very fast with the CPU. It will support the soft sector Z-89-37 type format. The All in one unit will have this feature, the kit versions to be released later will give a choice of 48 TPI or 96 TPI drives. Planned for early in '83 a 5" Winchester will be available to take the place of one of the internal 5" drives, giving the all in one unit over 5 megs built in disk storage!

We can expect initial sales by late summer or early fall, with the low profile units having video board fully populated with all 3 banks memory (for color), and the all in one with 1 bank memory (Black and White). A color monitor may be coming for the all in one as a built in unit. Because of additional time and hassle with FCC certification, expect kits to appear at least 3 months after assembled units are shipped. CP/M-85, HDOS and 8086 software will be available. As shipped, Condor and MBASIC will not run on these units from the Z89 because of the Z89 checking built in. Software said to be shipped with this unit will include BASIC-85, CP/M-85, and Z-DOS as the DOS, which is comparable to CP/M-85. Z-DOS was developed by Microsoft under the name MS-DOS, also known as IBM's PC-DOS (the DOS for the IBM personal computer. Since your present CP/M, or BASIC-80 will not run on the machines, you will have software that will. ZBASIC is similar to a beefed up version of MBASIC, but has extra commands for graphics and color control. The CP/M-85 will handle most of your present CP/M programs. A graphics demo disk will be shipped with this unit, and a utility to convert some of your H89 software to the Z series format. HDOS 3.0 to support this machine should be available late this year. It will include Hex, 0-org HDOS, and support the Z-67 type winchester drives on the H8 and H89. Micro Soft is doing lot of the software for this unit, and the IBM, including color software. Expect to see the H-P digital plotter and color plotter to follow for these machines, and a pixel resolution light pen perhaps in the future. The light pen capability is already built into the machine. There is a possibility of multi-user such as MPM-86 and maybe UNIX in 1983. Other business software being prepared includes electronic mail, business graphing, statistics, electronic card file, enhanced word processing and financial reporting packages. I understand the word processor will be a re-worked Wordstar.

Look for prices of between $4000 to $5000 for the Z120 all in one with 2 48 TPI drives and 128K memory.

All in all, a very impressive machine, and very well priced considering what you are getting. As usual, with Heath you get more built ins and less to buy to get the thing going and expand it. The Z machines are very versatile and very programmable. It appears as they have built in all that would fit, so the bare machine is actually quite complete. With the great speed of this unit, the expandability features, multitude of software, industry standard S-100 buss and CP/M operating system, high resolution graphics and oversized switching power supply, I doubt if you'll find a better machine for the money, and definitely not with the support Heath is planning. They will grow and be around for a long time. Zenith's marketing and name has been helping sell the Zenith systems. This has pushed Heath/ Zenith's computer sales up. Facilities are being enlarged and modernated with beefed up production. This will eventually mean better support and more products for that system anyway. Of course, a lot still depends on the top men, store managers, etc. We'll have to see how that turns out.

FEEDBACK, SIDE 2

I finally got around to looking at and evaluating side 2 of the H-SCOOP questionaires which were sent out in January. You already have the brief info on side one. Side two dealt mainly with comments related specifically to this newsletter. As I have said before, you can't please all the people all the time. I try to please the majority, that's the way I sell subscriptions. As an example, as many individuals who said they'd like to see more HDOS stuff and less CP/M, about an equal number wanted more CP/M and less HDOS. Likewise, one group wanted more H89 and less H8 info, and about an equal number wanted less H89 and more H8. Obviously the only solution to this would be to put out 4 newsletters. One dealing with HDOS for the H8, another for HDOS with the H89, another for CP/M with the H8 and yet another for CP/M with the H89. The second alternative would be to publish about a 50 page newsletter monthly, and then it would become a magazine, and not a newsletter. I want to bring these points out so you can sympathize with me in what I print and how I decide what to print. I try to please as many as possible while covering the greatest amount of material as possible. That was the reason for my "double density" format of this newsletter. NO OTHER newsletter of this size contains the number of words, but more important, the amount of information contained within these pages. I have done everything possible to get the most information into the smallest space, including using proportional spacing, less space between lines, printing a much larger copy and having the printer reduce it at printing, and having an extended issue when necessary to 'catch up'.

There were many questions about CP/M and HDOS also. I don't want to start that war all over again, but do want to make a few comments. I feel HDOS is a better operating system than CP/M for versatility, utilities available, friendliness, and ease of learning and use. I feel Heath did a great job with this Disk Operating System. Unfortunately, they threw it out there and said "here it is, do something with it". Much of the good software, especially when you get business minded, is under CP/M and not HDOS. This forces one to turn to CP/M, not because they like it, or want to use it, but it's that or forget the business software. That's it in a nutshell. Most of the response I have received agree with this. They like HDOS best, but had to go to CP/M because of the good and vast amounts of software available for it. Of course there are also those die hards who curse HDOS or CP/M, but they seem to be few. The way I look at it, the operating system allows you to bring in your program. Once it is in there, whether HDOS or CP/M, the DOS is fairly invisible and you really don't know or care what you are in. All you know is the program runs.

The information printed below represents the feelings, opinions and desires of the majority. As I hoped the majority of my subscribers like the newsletter just like it is, feeling it is a good balance of material and format, making information available which would not otherwise be available. A few got on my case for bad structure, usage and grammar, most noting the improvement of spelling over the years. The majority also like my telling it like it is, one saying "All other newsletters and magazines are too political - you tell it like it is - I will always be a subscriber". While some feel my editorials serve a purpose and should be continued, the majority want to see less editorializing, and would like to see the editorials more moderation and less editorials. All liked the editorial on Bulletin Boards in the February issue and felt it was a good example of what an editorial should be.

A lot of the novices want to see more information on a novice, or beginning level. Again, there's only so much you can do with one newsletter. I try to pick a balance of technical material, the nature of which is not too technical or too much on a novice level. I feel those who have been with me from the first issue received much of the novice background required, such as how to use PATCH and DUMP and these types of things. Naturally, new subscribers who do not receive the back issues miss out on this. It is impossible to continue to go into the detail for all info, or as I said earlier, I'd have a 50 page issue every month. I expect by now everybody is familiar with using PATCH, DUMP, and utilities required for doing good things, such as SUBMIT. Documentation is available for all of these, usually in the user manuals or doc files provided with the program. One cannot expect a newsletter to provide "how to" instructive material, that is provided elsewhere; or the newsletter would get mighty boring for most of the readers. I just take the time to point this out so you know where I am coming from and going to. If you are not familiar with these programs, get them out, read the doc and play with them. That's the only way you learn. Before I felt qualified to publish technical articles, publish a newsletter, give advice, or anything else, I spent hundreds of hours with my systems and much software and hardware to learn these things. "Go do likewise".

As I said when talking about side 1 of the questionaire, the majority want to see less details of NEW STUFF and less repeats. I have already starting cutting the NEW STUFF column to reflect only necessary and relevant info. I feel if you are interested you can contact the vendor, and I also feel it's the vendor's responsibility to keep you up to date on changes, updates, modifications, etc. If you wrote in to them or purchased their software or hardware, they have you on their mailing list (hopefully). It's now their responsibility to follow up.

I hope Heath and others take this next one to heart, and not lightly. Almost every one of my H8 subscribers said they felt that Heath had abandon them, and the H8 is either dead or dying. I feel along with Trionyx, D-G, Environmental Control Systems, MI-8 and yes, Heath and others that this is not quite true. What is happening is the H89 stuff has gained much more momentum, but there is still excellent H8 support out there, and there is more coming all the time. Many are wising up and making interchangeable software and hardware. Anyone wanting to make bucks, this is the market. GOOD H8 hardware. And yes, I do feel the cassette and H9 are dead. I receive less than 1 percent of my mail relating to these, and most of that is asking "why not more on them".

There was a much larger majority than I had anticipated that want to see more evaluations. Evaluations of business software, other software and hardware. I guess it makes sense that an evaluation is more important than a NEW STUFF announcement. I had always told vendors that a report of their stuff will sell a lot more items (if the report is positive) than a NEW STUFF mention will. Quite a few said they wanted to know more of the availability of software and hardware. I print all I get sent in to me, and then some. It's hard for me to visualize how to print more on software and hardware availability, and yet cut the NEW STUFF column. Seems to either be magic, or a bit of contradiction there. I guess the point is subscribers want to see more technical info, programming goodies, hints and tricks.

What else is wanted? **MORE!** More software routines, more discussions of languages and operating systems, more on hardware comparisons and compatibilities, more tutorial articles, more GOOD & BAD GUYS, more on the internals of MBASIC and HDOS, more projects and program listings, more on repair tips and "how I did this" type articles, more hardware and software tricks and tips, more maintenance and troubleshooting tips, and in general, more of H-SCOOP and more frequent publications.

It's enough of a-job to get this newsletter out once a month. There's an awful lot of "behind the scenes" stuff that is done that you would not even begin to imagine unless you are in the business. I will continue my policy of trying to cram as much information in each issue as possible and printing larger issues to catch up as needed. Only one person said they want to see MORE space between lines, the rest felt the eyestrain was worth it. And I admit because of the double density format it is harder to read, but it was my answer to give you more for less money.

Just to give you an idea of what has happened in the past few years, in case you have not been there, bear with me for a moment. When I started in this, there was only the H8 (which I still feel is the best general purpose computer available), the H9 video terminal, the H17 single density single sided disk drive and cassettes. This was not hard to support and please everyone. They all ran HDOS at that time and BH BASIC. Look at it now, just a few short years later, and it's expanding exponentially each year. How do you cover it all? How do you satisfy everybody? That is one reason I started out with H8 only. I did not want to bore you with ET 3400 or H11 stuff. Then the H89 came and since they used the same software and much of the same hardware such as disk drives and memory, I decided to cover that as well - still in the same family. Now we have 2 computers with another on the way, an unbelievable number of various disk controllers and drives and formats; HDOS, CP/M, UCSD PASCAL operating systems, cassette, a multitude of languages. . . Wait till you see what's coming from Heath and others in the next year!!! I just want you to understand what is happening here, where we came from, and where we are going, so you understand my position as editor, writer, publisher. . . And as usual, most want to see more, but the input from subscribers on things such as "how I did this" is minimum. Much of the info must be drawn from my experience. Don't let this or any other newsletter you support down. Only as you contribute can we continue to print and publish info to present to the world.

Anyway now that I know more of what you want, your gripes, comments, etc. I will try to go that way - where possible. I expect you to keep in closer touch with me. What are you doing? Anything unique. Tips? Problems? How did you fix it? Mods? What did you do? How did it work? Keep it coming. I thank all of you who took to time to respond to the questionaire. It shows you are the ones who care, and I very highly value your opinion. I wouldn't take the time to make up the questionaires and evaluate them if I did not.

On a final note, I am happy to report that the vast majority appreciate and support all the offers; hardware and software I have made available through this newsletter and its parent company, QUIKDATA, INC. The support and help I give above and beyond this newsletter is also appreciated, and it will all continue. Any time I come across some good hardware or software, such as the Tandon drives, Spellbinder, DBASE II, etc., if no one else is doing anything with it I will become a dealer and give you the best possible price and support on it, and I'm proud to say, thus far NO ONE has been able to top that. The reason for that is it is not my main line of work. It works out quite nicely with the newsletter and the selling end working together. That allows me to sell products with a very minimum markup, usually 10 percent or less. Oh well, on to bigger and better things.

WHO'S WHO

Last month I printed the address of K. Hatcher incorrectly in the Who's Who section. The correct address is 4735 Cowell Blvd #88, the rest was OK.

Ken Lind/ 510 PSC/ APO NY 09221 would like to associate with other Heath users in Italy or Germany. He also would like to swap software (not copyrighted stuff, and most in MBASIC HDOS). He has much software, not extremely advanced like word processors or DBM stuff. Most of the stuff he has was either developed by himself or obtained through a swap with others who developed it .

Carrol Waddell/ 114 Dorcel/ Sumter, SC 29150/ (803) 773-4076 built a parallel interface for the H8 to an Epson MX-80 printer. It only costs about $15 and does not require the H8-4 board or a serial interface on the printer. The only problem is that the BIOS in CP/M must be modified to use it. He will sell BIOS mod and schematics for $10.

Charles Juvenal/ 6923 'B' St/ Omaha, NE 68106/ (402) 392-2796 is interested in starting a newsletter. This one would be dedicated to reviews of advertised software. Charles has been burned many times by vendor claims in their advertising that have not proven true of software he has purchased. He therefore feels a need for this type of newsletter, featuring reviews. I think it is an excellent idea, and would like it to be expanded to hardware as well. From the feedback I received in my returned questionaires, this type of service is needed. Although I have been trying in this newsletter to cover some reviews, the space is just not there to do it. I think such a newsletter would be beneficial to all. If this sounds good to you, write to Charles.

Robert Iacullo/ 352 N Main St/ Doylestown, PA 18901/ (215) 348-3621 has been doing work with the DEC TU-58. He is interfacing software with the TU-58, which is a tape unit. It uses a standard serial port, and Robert is writing driver routines. If anyone has an interest in this area, contact Robert.

Bill Hall/ 5621 Maple Heights Ct/ Pittsburgh, PA 15232 is working on interfacing a CPT 4200 Selectric typewriter to an H8 via the H-8-2 parallel port. He is also writing ASM programs for driving the selectric. Anyone interested, contact Bill for free assistance.

REQUESTS

Steve Haggins/ POB 1260/ Enterprise, AK 36331 wants to know if anybody knows where the system date is stored for the Heath UCSD Pascal system. He would like to write a routine which would display the date, or be able to call the date from this Pascal.

Rodney Nelson/ 2231 Ablee St/ Eureka, CA 95501/ (707) 443-9809 has been having problems with his H89 display. If this sounds familiar to anyone, give Rod a call and bail him out. His H89 will randomly change or add characters after they are displayed on the screen. Originally, the graphics would fall apart during a game and Heath recommended changing U467, 468, 469, 470. This seemed to solve the graphics problem. It was then noticed that characters were changing and being added between words at any time where before there was no problem. Heath told him to replace U424 and U425, but to no avail. If Rod uses PIE for editing, and characters change on the screen text, he can scroll up off the top and back down and the letters revert to the correct characters. Any help on this one, write Rod.

Y. Lee/ 1824 Save Ct/ Timonium, MD 21093 is trying to use MODEM7 from the SIG/M library (volume 48). To patch the program for the Heath systems, he needs to know 1) MODEM status port, 2) Bit for checking modem "ready to receive" and 3) Bit for checking modem "ready to send". If anyone is familiar with this modem package, please contact Mr. Lee and bail him out.

PROGRAMMING GOODIES

CONTROL OF H8 FPLEDS from MBASIC

Leonard Simonson wrote this one after the request several months ago for controlling the H8 front panel display from MBASIC.

POKE 8220,129 will turn on the display for monitoring registers or memory. I want to add to this, that doing so will slow down your program somewhat, since processor time is spent updating the display. **POKE 8199,2:POKE 8197,N** can be used to display the contents of a register, where N=0 for SP, 2 for AF, 4 for BC, 6 for DE, 8 for HL, and 10 for PC.

POKE 8199,1 Display memory location.
POKE 8212,N where N= low byte of memory location to display
POKE 8213,N where N= high byte. In the following example, a sequence is presented to turn on the display, and display the contents of memory location 040.100. Note that the letter O follows the '&' symbol to show you are working with split octal numbers.

```
POKE 8220,129:POKE 8199,1:POKE 8213,&O040
POKE 8212,&O100
```

This conversion of split octal feature can also be used as in the following example. Convert 040.100 split octal to a decimal number:

```
PRINT &O040*256 + &O100
```

Another useful command is turning on the display for user update. This is accomplished by the following:

```
POKE 8200,131
```

Try this example:

```
10 POKE 8200,131
20 FOR A=3 TO 9 STEP 3
30 POKE 8200+A,146
40 POKE 8201+A,144
50 POKE 8202+A,255
60 NEXT:END
```

Memory locations 8203 thru 8211 hold the contents of the front panel LED's. The LED segment is made as follows:

```
    1
 6     2
    0
 5     3
    4     7
```

The Display Bit Pattern is: 7 6 5 4 3 2 1 0

```
POKE 8203,0       All segments lit
POKE 8203,255     All segments off
POKE 8203,141     Prints "C"
POKE 8203,&O215   Same, using Octal
POKE 8203,&H8D    Same using Hex
```

To turn the display off, **POKE 8200,193**

These are a few sample things you can do. Play around with it to get the feel for what you are doing, and you will open a whole new world for the H8.

CP/M MULTI PRINTER CONTROL

Unlike HDOS where you can drive many printer devices simply by assigning a different name such as AT: and LP:, or LP1:, LP2:, etc, CP/M outputs normally to the LST: device. It is more difficult in CP/M to drive various printers. One easy way most CP/M operating systems allow you to change the I/O byte, which is at location 3, by using the STAT command. This allows you to change which port the LST device will be. When you boot up, assume you are at port 340 for LST: which is LPT device. Now assume you have TTY set up for port 320 and a different baud. By executing the following command, LST will now assume port 320 as the TTY address.

STAT LST:=TTY:

Other CP/M's may handle it differently. Consult your manual under STAT command for your particular system. Another way pointed out by Doc Campbell and Caroll Waddell, among others, is to manually set the I/O byte. This can easily be done from inside an MBASIC program for instance. By knowing what this byte at location represents, and again, consult your manual, you can use a POKE instruction, or whatever, to change it to a different value. This value represents what device will become the active LST: output device.

TECHNICAL FORUM

DOWNLOADING LARGE CP/M FILES TO SMALL DISKS

* Have you wanted to buy CP/M software, such as **Word**, only to find it isn't available on a 5" DS DT disk for your Tandon 100-4? Also one of its files is 136K, too long for the 5" SS SD disks of your H89. It comes only on an 8" SS SD IBM format disk. To make matters worse, no one handy has a system with both the Tandon 100-4 and an 8" drive attached to the same H/Z89. Here's how to do it if you do have a friend with an 8" drive connected to his H/Z89.

Examine the directory of the distribution disk with any of the several disk utilities available. The HUG utility SDUMP.COM will access and EDIT files in almost any CP/M disk format (Elektrokonsult's DDUMP also does an excellent job - ed). All of the directory of WORD will be found in Group 0 of the distribution disk (the DIRectory of a CP/M disc will be found somewhere in the vicinity of track 2 - ed). The long file MAINDICT.CMP, will be listed in nine entries in the directory. Note the number of the Track and Sectors in which all nine Extents are entered so individual sectors can be recalled for EDIT.

When a file consists of more than 5.625 Extents it cannot be copied, as is, to a 5" SS SD 40 track disk.

Note that each entry in the directory consists of 32 bytes in Hex, i.e. two lines as displayed. The first byte of active files will be 00 while the first byte of those which recently have been erased will be E5. The next 11 bytes contain ASCII code for FILENAME.EXT. 20's fill the bytes reserved for letters not used in naming the file. The 13th byte contains the number of the last Extent of that entry, starting with 00 (an EXTENT is a 16K chunk of disk space, or 128 records with each record 1/8 K - ed). The 16th, or last byte on the first line, gives the number of Records in the last Extent of that entry (as a hexidecimal number). If the Extent is full, the sixteenth byte will be 80H, i.e. 128 Records.

Each directory entry of both 5" and 8" SS SD disks can only address one 16K Extent as it consists of Groups of 1K bytes. Since both Heath and MMS versions of CP/M write files on 5" DS 80 track disks as Groups of 2K bytes, two Extents, totaling 32K, can be listed on one line of each entry in their directories. If a file longer than 90K bytes is to be transported on several 5" SS SD 40 track disks, it must be divided into sections. Each, but the last, of the divided files should be four Extents or 64K bytes long. Then they can be copied from 5" disks to larger disks of any format. (If several double density 5 inch disks are to be used to transport a very large file, it can be divided into larger sections, each of an even number of Extents.)

Dump the sector of the directory of the distribution disk (or a copy of it if your friend has means to make one) listing the Extents of MAINDICT.CMP starting with 04 in EDIT mode of the utility used. A New file is made by changing two bytes in each file of Extents 04 through 08. Change the last letter of the name of the extension, say from 'M' to 'N' (4D to 4E), and change the number of its Extent, in the thirteenth byte, from 04 to 00. Make the same change in the name of all Extents of successive entries and give consecutive numbers to the Extents from 00 thru 04 (04 only because the rest of the file can fit on the last 5" disk, otherwise the last Extent would be 03). This breaks the large named file into two smaller files each with a different name, to be re-assembled later on the tandon drive -ed.

After these changes have been made in the directory of the distribution disk, the much shortened original file of four Extents and the new file of five+ Extents can each be PIPed to its own 5" SS SD 40 track disk. V at the end of the PIP command provides validation. Once you have the copies, you may want to use SDUMP again to reverse the changes made to the source disk.

The next step is to copy each of the renamed files from the two 5" SS ST disks to one 5" DS DT disk on your own H89. Just be certain to transfer files in the same order as they were found on the source disk. After copying is completed use SDUMP in the EDIT mode again to reunite the two files back into one on the new destination disk. Change the file extensions so all again are the same as they were originally. Also be sure to renumber the Extents back into one consecutive chain; i.e. 01, 03, 05, 07 and 08 on a disk with 2K Groups. Should directory entries appear which have 00 Records and 00's for Group numbers, change the first byte ahead of the FILENAME from 00 to E5 so CP/M can use that directory line for another file entry.

Many thanks for this handy one from Lloyd E. Johnson 700 Highview Rd. East Peoria, IL 61611. You may not need it now, but when you do, you'll be glad you know where to find it.

Just a little info to help you with this, as to how CP/M organizes a disk. Normally the first 2 tracks for 8" or 3 tracks for 5" discs is reserved by CP/M with the bootstrap loader on track 0 sector 1. The CCP is located on the next 16 sectors, with the BDOS following on the next 28. The BIOS follows on the final 7 sectors. This is a general rule, and is dependent on the drives, format and density being used. The disk directory follows. For 8" DD DS drives in the Magnolia format, that happens to be at track 3 and sector 1. Depending again on the CP/M and system used, the directory can accept anywhere between 64 and 256 entries.

MORE READER TECH FEEDBACK

Clif Lesperance called to tell me that another cause of system crashes can be **tarnished IC pins.** I'll go one further and include tarnished buss connector pins. This was covered in previous Tech Forum articles on H8 system crashes. Some IC pins and buss pins clearly display dark tarnishing or oxidation. This will have to be removed some way to make good contact.

There appears to be a **bridge rectifier** and **voltage regulator problem** on some of the H89's, especially the earlier ones. The symptoms start with dark bands appearing on the screen and moving slowly. This is most evident if you cause the screen to be painted in reverse video, or if it is filled with graphics. It seems that under voltage causes this, and in many cases the bridge rectifier will either blow or shut down sooner or later. Erratic CPU operation may also be observed if the supply drops low enough. Heath is now using a new bridge rectifier that solves that problem. I had installed a 25 amp bridge which I purchased from Radio Shack (part 276-1185). This was also found by William Rousseau, who substituted the same Radio Shack part, and he is still having problems, but they seem to be related to other things.

The 5 volt regulator is another cause for problems. The original H89 as supplied had a lighter regulator. If you install the Magnolia DD board you get a new heavy duty regulator and heat sink, and I understand the H90's also give you this. The present voltage regulator cannot put out enough amperage to handle extra boards, memory, etc that are filling up the H89's. A part to substitute would be the LM323 or 78H05, at U101. This is a 3 amp unit to replace the present 1 amp unit. That with the bridge rectifier change should give you plenty of juice!

Another item reported was heat build up in the H89. One individual cut out every other slot on the top of the H89 to allow better air flow. Of course you can also disconnect the fan and operate with top off. As to heat dissipation, I guess the H90 upgrade also includes heat sinks for all the voltage regulators on the power supply board.

A short note on thermal runaway. Usually a bridge rectifier and voltage regulator will "shut down" if too much current is drawn from them causing them to over heat. When allowed to cool, they work as normal till they heat up again. Thus if you have problems when the machine is running for awhile, and cooling it down temporarily solves your problems, then suspect the regulators and/or bridge rectifier components. This feature of shut down is a safety feature built into the devices. Instead of shorting and allowing too much voltage to pass through, they simply shut down, or turn off. Of course, if the current draw is excessive, they will probably blow up and all the cooling in the world will not help them any more. It's always a good idea to stock one of each regulator in use and a bridge rectifier. The 3 pin flat 5V jobs are a must to stock, since you see them all over the place. These are 1 amp units. You also may want to check the output voltages. I've seen some as low as 4.2V and as high as 5.5. The 5.5 is getting close to being too high, but the 4.2 is definitely too low. It never hurts to carry an extra fuse or two, or in an emergency a #8 wire will suffice (just kidding!!!!).

Feedback from William Rousseau on his voltage problem. He thinks it is narrowed down to a transformer problem, probably supplying too much voltage. He said pin 5 on P101 is charred, and this is an AC input to the bridge rectifier. He also said that it appears that the MTR-90 conversion draws more current than the MTR-89 setup, as his systems with the MTR-89 seem to run cooler than the MTR-90 one. Another problem William passes on was when swapping CPU boards he noticed his screen characters suddenly became ripply. The boards looked OK. Suddenly the screen went black and he heard a snapping noise. He found the anode lead from the flyback transformer came in contact with and was arcing to the ground wire (pin 2) of the cable assembly connecting to P514 of the CPU board. By rearranging the cable and anode lead, all was well again. **WARNING:** The anode lead carries very high voltage. Stay away from it. Never work around that area with power on or line cord plugged in. Even with the unit OFF and unplugged, the anode lead to the CRT can carry many thousands of volts, enough to knock you on your rear, and cause you to drop anything (like an H89 or H19) you may be holding!

Interestingly, Frank O'Neal, Senior Consultant at Heath writes to Dr. Rousseau, "The problems you described in your letter are just now being realized as a problem rather than isolated failures. From the information you supplied in your letter it sounds as though you are on the right track. The 25 amp bridge rectifier you installed in your H89 will probably accomplish more in correcting the problem than any of the other solutions you suggested. . ."

David Orsz sent in feedback on the Heath Z-H8 CPU card. He added a ON-ON switch near the top of the board and wired it to the pins for 2 or 4 Meg booting on Power up. This allows him to easily select between 2 or 4 MHz operation. I have done this on my H89 mods and mounted the switch on the rear so I can easily go from 2 or 4 Mhz--ed.

Charles Loeb wrote in a handy idea for the H89 using 2 disk controllers. The H89 has a covered opening for a 50 pin socket for the Heath 8" drives. The additional 34 pin socket can be installed by cutting the removable cover plate with a sheet metal snips and drilling a small hole to attach between the connector and the second hole on the 50 pin cutout.

I have a Z89 and that did not have the metal plate, but a pop off plastic cover. I used an oversize washer on the 34 pin plug connector (male) to mount it in the 50 pin hole. What about 3 drive cables? Well I mounted the 50 pin plug in the 50 pin hole, the 34 in the 34, and made a cable with the 34 pin plug which just hangs out the back. This way I can quickly change or plug any combination without opening the system. I did the same thing on my H8's both for the RS-232 connectors and the disk drive cables. All are now mounted on the back. I very seldom open up the H8 anymore except to change or adjust boards-- ed.

H89 ROM's AND THINGS
or
ALL YOU WANTED TO KNOW ABOUT PART #'S

Many individuals have been calling with confusion about all the different ROM's and parts used in the H89. I will discuss several of the more important parts.

MEMORY DECODE ROM. This is located at position U517 on the CPU board and can be one of two parts. 444-42 and 444-66. The first was the 'old' part and was used for 48K or less RAM. This part number does not permit the use of more than 48K RAM, nor was 0-org, thus CP/M would not run either. The new part, 444-66 allows 0-org CP/M to be run, and allows up to 64K RAM addressing. This is the one recommended for ALL systems. Using this ROM, memory selection is accomplished with jumpers JJ501, JJ502 and JJ504 as follows. For 16K, these should be 0, 0, and 0 (or B) respectively. 32K would be 1, 0, and 0 (or B). 48K would be 0, 1, and 0 (or B), and 64K, which requires an additional 16 memory board to be installed, should be 1, 1, and 0 (or B). New CPU boards only have JJ501, JJ502, and JJ503 and follow the same order as above, only substitute JJ503 for JJ504.

I/O DECODE ROM. This is located at U550 on the CPU board either as 444-43 or 444-61. 444-43 is the old number which supports cassette port, serial I/O card and H17 disk drive card. Part 444-61 supports two separate disk cards, and the three port serial I/O card, but no longer the cassette port.

MONITOR CODE ROM. This is located at U518 and is available as 444-40 for the MTR-88, 444-62 for the MTR-89 and 444-84 for the MTR-90. The MTR-88 version is only used with cassette tape and the H17 controller. The MTR-89 version supports the H17 controller and the H-47 dual 8" controller, but not the cassette tape. The MTR-90 version supports all disk devices, and is supplied with the double density H37 controller, and the hard disk controller. There is more code involved with this, thus it is a 4K ROM, while the first two are 2K ROMs. Because of this, if the MTR-90 is installed, the **SECONDARY ADDRESS DECODER**, located at U516 must also be changed.

SECONDARY ADDRESS DECODER has two part numbers, 444-41 for use with the MTR-88 and MTR-89 monitor ROM, and the 444-83 for use with the MTR-90 monitor ROM. Associated with the Monitor code ROM and secondary address decoder are jumpers JJ505, JJ506, JJ507 and JJ508 for older units, and JJ504 through JJ507 for newer units. The MTR-88 and MTR-89 require the jumping to be 0, 0, 0, 1 (or B) respectively, while the MTR-90 requires jumping to be 1, x, 1, 1 (or b). The x indicates a jumper to be installed between center pin of JJ506 (or JJ505 depending on which unit you have) and pin 14 of P508.

MISCELLANEOUS BITS

This is all that is necessary for a CP/M system. If you are using the system for HDOS, additional parts must be installed. A special HDOS ROM must be installed at U520. Two 2114 RAM's (1K X 4, P/N 443-764) are needed at U523 and U525. This is the floppy scratchpad memory. In addition, if a drive is installed in the H88 or H89, a 12 volt 1 amp regulator must be installed at U103, 78H12 (P/N 442-650). If the 16K memory expansion board is used, you must change U562 from an 74LS132 to a 74S132. This is usually supplied with the memory expansion board. Current H89's use a 3 amp 5 V regulator at U101 to handle all the extra boards and parts installed in the H89. This is part 78H05. Older units used a 1 amp 5V regulator LM309K. New units also now have the regulators U101 to U103 mounted with heat sinks to help dissipate the heat for greater output capacity. It probably is a good idea to do this with all units. Is appears that newer H89's have a heavy duty bridge rectifier. The old H89's have a regulator that sometimes does not stand up to the load, and has been known to cause major problems. Refer to above section on Reader Feedback.

DRIVE PROGRAMMING

A real confusing issue seems to be drive shunt programming. It is confusing because Heath choose a nonstandard way of addressing drives. Magnolia uses the standard, and the H37 uses the standard. Briefly, only one drive in a chain should have the terminating resistor pack installed, and although recommended to be the last drive on the chain, it's really not that critical, as long as one drive, and only one drive has it installed. On all drives, the MX (multiplex) shunt should be open, this is a must if operating more than one drive. Wangco or Siemens type drives use a head load solenoid and must have either HS or HM closed. If HS is closed, the head will be loaded ONLY when that particular drive is selected. If HM is closed the drive head will be loaded whenever ANY of the drives are selected, since it is loaded with the motor on signal, and this is common to all drives. The TANDON variety do not use head load solenoids, the head is loaded whenever the door is closed, thus it is not necessary to close either HS or HM. As a reminder, never close the door of a Tandon type drive without diskette inserted as you'll slam the heads together.

Now for the confusion. Some drive select lines are labeled NDS0, NDS1, NDS2, and NDS3 as is true for the Tandons. Others such as the Wangco or Siemens have DS1, DS2, and DS3. However these are labeled, they apply to select drive 1, select drive 2, etc. Standard procedure is to close select drive 1 (NDS0 or DS1) to have that drive select for drive number 1 (SY0: for HDOS or A: FOR CP/M). In a similar maner, NDS1 or DS2 is closed for selecting as the 2nd drive, etc. This is true for the Magnolia card and the H37 card and any other card that uses standard addressing. Heath does it differently. NDS2 or DS3 is closed for drive select #1, NDS1 or DS2 for drive select #2, and NDS0 or DS1 closed for drive select #3. Under no circumstances should more than one select line be closed on each drive, and each drive should not have any more than 2 shunt plugs closed totally. I find the easiest thing to do is remove the shunt plug and 1) either replace with dip switch, which is best for lot of drive switching or experimenting, or 2) insert short wires across shunt for line you want to close.

I hope this has helped you out of some of the confusion of configuration and parts. Keep this handy for future reference.

Sources for this information were obtained from Charles Bennett, Heath Manuals, Drive manuals, HUG BB and other varied sources. Many thanks to all.

HDOS FLOPPY DISK ERROR HANDLING

I. SECTOR FORMATS

Note: This article was written by Terry Smedley, and appeared on my HS-012 Disk Drive Exerciser software disk. Since we all experience errors under HDOS, and many different types, using various STAT programs, I felt it beneficial to make this information available for your general curiosity. When Terry refers to the program XSTAT.ABS in the article, this is a comprehensive STAT reporter program which was also contained on disk HS-012. For those interested in this disk, it is available from H-SCOOP for $18 as written up on OFFERS in June issue of H-SCOOP.

All Heath-formatted disks (HDOS, CP/M, UCSD) used in the H17 disk system have sectors written in the following format:

Beginning of Sector (index hole)

HEADER:

```
        10 zero bytes
*       sync character (375Q, 0FDH)
*       volume number
*       track number
        sector number
*       header checksum
        zero byte
```

DATA:

```
        10 zero bytes
*       sync character
        256 data bytes
*       data checksum
        data checksum
        data checksum
        data checksum
```

Originally the three extra checksums as seen at the end of the DATA were written to insure the tunnel erase gap had passed.

(The asterisks in the left hand column indicate the error-types which are logged by the Heath Disk Operating System.)

The sync characters are used by the H17 software/controller to locate the exact spot where data begins in the sector.

Checksums are computed as follows:

assume: the checksum before reading or writing the first byte is zero

(A) contains the nth byte (read or written)

(D) contains the checksum after the (n-1)th byte

then the checksum for the nth byte is computed by

```
XRA   D     compute difference bits
RLC         rotate left one bit
MOV   D,A   update checksum for next byte
```

This checksum is computed as data is read from the disk and compared to the checksum which was computed in the same way when the data was written to the disk. If the two checksums differ, one or more bytes in the record are in error and a checksum error is flagged. Note that two separate checksums are written in each sector: one for the sector "header" (volume, track, sector) and one for the 256 byte data block.

II. ERROR TYPES

Disk errors in HDOS are generally categorized as "soft" or "hard". Whenever a disk error is encountered, HDOS attempts to re-read the data up to ten times. If the data cannot be properly read after all 10 REtries (11 attempts total), the error is classed as "hard". All other errors are considered "soft" or "recoverable". HDOS maintains counters for both soft and hard errors. In addition to these generalized soft and hard error counters, six distinct error types are recognized and counted.

The first four disk error types (those in the HEADER section) are trapped in the D.LPS (Locate Proper Sector) routine. Before any read or write operation, D.LPS is called to position the head over the proper sector. D.LPS will trap the following errors:

MISSING HEADER SYNC Couldn't locate the sync character preceeding the header data. The H17 ROM routine R.WSC is called to wait for up to 25 character times while searching for the sync character. If the sync character is not detected in this time, the missing sync error is flagged. Note that under normal conditions the sync character should be detected within 10 character times.

WRONG VOLUME NUMBER The volume number in the sector header is not the same as the volume number recorded as being currently mounted on the drive. The two-byte word at D.VOLPT (040.247) points to the memory location containing the currently mounted volume number.

BAD TRACK SEEK The track number in the sector header is not the same as the target track for this operation. The target track is found in memory location D.TT (040.240).

HEADER CHECKSUM ERROR The checksum computed while reading the sector header does not agree with the checksum written on the disk.

These four errors may be flagged either during read or write operations. The remaining two errors are detected by the routine D.READ only during disk read operations. Note that there is no special error type flagged if the sector number in the sector header is not the same as the target sector for the current operation (found in D.TS at 040.241). LPS allows the disk to make two full revolutions (passing over 20 sectors) while searching for the proper sector. If the proper sector is not located within this time, a soft error is recorded, but no special error type is flagged.

These two errors are flagged by the routine D.READ:

MISSING DATA SYNC Essentially the same as the Missing Header Sync error described above, except that it is the pre-data sync character which was not located.

DATA CHECKSUM ERROR The checksum computed for the 256 byte data block differs from the checksum on the disk.

III. ERROR RECORDING

Two routines in the H17 device driver are responsible for keeping track of disk errors. D.ERRT updates the error-type counters. A separate counter (one byte) is maintained for each of the six error types described above. When one of these errors occurs, D.ERRT is called to update the appropriate error counter. These error counters are found in the six bytes beginning at D.E.MDS (040.265). Refer to the EDRAM.LST file for details on the H17 RAM cells. A second routine, D.CDE (Count Disk Errors) is called to update the generalized "soft" or "hard" error counter and to re-seek the target track if necessary. The hard error counter is a single byte at D.HECNT (040.261): the soft error counter is a two-byte area at D.SECNT (040.262). An error is considered "hard" if the data cannot be read correctly after 10 REtries (11 attempts total). All other errors are "soft" or recoverable errors. The byte at D.OECNT (Operation Error Count, 040.264) is initialized to 10 by the D.SDP (Set Disk Parameter) routine before each read or write operation. Each entry to D.CDE decrements this location by 1. When D.OECNT is decremented from 0 to -1, a hard error is flagged.

XSTAT will display the number of disk reads and writes as well as hard and soft errors which have occurred since the last system RESET. If any errors (hard or soft) are found, then the contents of the six error-type counters are also displayed. If you are experiencing frequent disk errors, then knowing which type of error is occurring may help you locate the problem. It is important to consider the ORDER in which the disk errors are flagged when attempting to diagnose a problem. For example, poor stepper motor operation results in some seeks where the head is slightly off the center of the target track, the error that is likely to be flagged is Missing Header Sync. The LPS routine will fail to properly read the sector header sync character if the head is not in position directly over the written track. The disk errors are flagged in the order indicated in the sector format description (I. above).

If any hard errors have occurred since the last time XSTAT was run, the track and sector where the hard error occurred will be displayed. This information may help you locate a bad sector on a diskette, or a particular track which your disk drive has trouble seeking. XSTAT clears the error track/sector after it has been displayed.

IV. FACTORS AFFECTING THE HEATH SY: DRIVER ERROR COUNTS

Because of problems with the way the Heath SY: driver records soft errors, you should consider the soft error count as an "estimate" if you are using this driver. The UltiMeth (HUG) SY: driver counts errors precisely; the soft error count will be reliable with this improved device driver. The Heath SY: driver uses a D.ERRT routine which increments the soft error count (D.SECNT) as well as the error-type count. This results in many (but NOT ALL) errors being counted as two soft errors. The HDOS 1.6 STAT command divided the soft error count buy two before display to correct for this double counting. XSTAT displays the actual number of soft errors recorded, so the number of soft errors reported by XSTAT will be different than the number reported by the HDOS 1.6 STAT command recorded.

To see why the Heath SY: soft counts may be inaccurate, consider the following:

The LPS routine allows 20 sectors to pass while attempting to locate the target sector. All errors encountered while reading (up to) 20 sectors will be counted by D.ERRT. If the target sector is not found after this two revolution search, then a (perhaps additional) soft error is recorded by D.CDE. Then, the head is repositioned over track 0, moved back to the target track, and the LPS routine is entered again. This process is repeated for up to 10 retries (11 entries to LPS total).

Take a hypothetical case of a drive which hangs up between tracks 6 and 7. Assume a read command has been issued for track 7, sector 0. When LPS is called, it will probably fail to locate the Header Sync Character on each one of the 20 sectors it attempts to read. This will cause the soft error count (and the Header Sync error count) to be incremented to 20. When LPS returns to the D.READ routine (with the Carry flag set to indicate that the desired sector was not found), D.CDE will be called to indicate an additional soft error. If you assume that the read/write head hangs up in the same spot on each of the 10 re-entries into LPS, then the error count (just before a hard error is flagged) will be:

(11 entries to LPS) x (20 errors/entry in LPS) + 11 errors (D.CDE) = 231 soft errors

When D.CDE is called after the 10th retry, a hard error will be recorded. D.CDE attempts to adjust the soft error count by subtracting ten from the low order byte of the soft count. The Heath code assumes improperly that there will be precisely 10 soft errors for each hard error, and that no carry (borrow) will occur between the high and low byte of the soft error count when the correction is made. Thus, at the end of this aborted read operation, the final error count will be:
Hard Errors: 1, Soft Errors: 221

The HDOS 1.6 STAT command would show a soft error count of 110. This is far from the desired count, which is 1 Hard Error and NO Soft (recovered) Errors. The counter for Missing Header Sync would correctly show 220 (11 entries x 10 errors/entry). Admittedly, I have chosen a "worst-case" analysis, but similar considerations for other, more likely, error types will show why the Heath soft error count is at best approximate.

V. ERROR COUNTS WITH THE ULTIMETH DRIVER

The UltiMeth driver will properly report hard and soft errors for the case described above. This driver initializes a "soft error seen" flag when D.SDP (Set Disk Parameters) is called from each READ or WRITE entry. When the first soft error is recorded, this flag is set to indicate that an error has already been found during this operation. Subsequent errors for the same operation will not increase the soft error count. The error type counters will continue to be updated. If, after 10 retries, the data has still not been properly read, the soft error count is decremented (by 1) to remove the soft error, and the hard error count in incremented. This procedure insures reliable error recording under all conditions.

VI. ERROR COUNT PHILOSOPHIES

Handling of disk errors which occur within the LPS routing requires a design decision. Should every error detected in LPS be counted as a soft error even though such an error could occur as many as twenty times for each entry to LPS? The Heath SY: includes individual errors within LPS in the soft error count. This can result in substantial over statement of the error counts (see the discussion in IV). An alternative philosophy is to record a soft error only if LPS cannot locate the target sector within the 20 sector search allowance. This eliminates the over statement problem, and in fact is the approach used by Heath in their implementation of CP/M. Individual error-type counters should continue to be updated for all errors, regardless of whether LPS eventually finds the target sector. Another decision must be made regarding recovered error counting. If a particular error is recovered after 6 retries, should this be counted as 1 recovered error or six errors? The Heath SY: driver counts this as six errors; the UltiMeth driver records only 1 soft error. The Heath count (in this circumstance) is probably more reflective of actual system performance. A large number of retries required to recover from errors indicates marginal system performance. The UltiMeth error count may tend to mask this effect by counting only 1 error regardless of whether it took 1 or 10 retries to recover. From the author's point of view, the most consistent approach to error counting is a mixture of the Heath and UltiMeth approaches. Soft errors occurring within the LPS routine should be recorded only if LPS exits without locating the target sector. Individual error-type counts should be updated for all errors. Each attempt to recover from an error should be counted as a soft error. This philosophy requires the soft error count to be reduced by the retry count (normally 10) when a hard error is flagged. This also allows numerous "sub-soft" errors to be recorded in the error-type counters without a soft error being recorded, if errors occurred within LPS that did not prevent this routine from locating the target sector. XSTAT would have to be modified (slightly) to allow display of the error-type counters even if no soft or hard errors are found if this philosophy is to be implemented.

VII. A DISK ERROR EXPERIMENT

You can demonstrate the difference in error handling between these two drivers very easily. Mount a disk in both SY0: and SY1:. The disk mounted in SY0: must have XSTAT.ABS on it. Run XSTAT to determine the current number of errors recorded. Now, WITHOUT Resetting or Dismounting/Mounting, insert another disk (with a different volume number) into SY1:. However, the volume number of the new disk will not match the volume number in the drive table and the LPS routine will flag a wrong volume number error for every sector read (20 sectors x 11 entries to LPS). You will hear the head move to track zero and then reseek the desired track before each reentry into LPS. When the operation completes, and an error is flagged on the console, run XSTAT. The UltiMeth driver will show 1 Hard Error, 0 Soft Errors, and 220 Wrong Volume Number errors (assuming all counters were zero when you started). The Heath driver will show 1 Hard Error, 221 Soft Errors, and 220 Wrong Volume Number errors.

GOOD GUYS and BAD GUYS

GOOD GUYS--Dr Jay Gold--Anton Ley purchased his package and was very happy with it (Home Finance System). He wanted some changes and called the doctor who took the time to give the info needed plus good advice to boot. With his help, Anton got the results he wanted. I have received similar reports from several individuals who remarked about Dr Gold's good service and excellent software.

GOOD GUYS--Ted Benglen of MI-8, for products available and outstanding support- -Roger Fennema. Roger also has good words for **EVRYWARE.** He had a problem with their Galactic Warrior game and wrote them a letter. He received a fast reply and was sent another disk. **KEYBOARD STUDIO** has also been a very good vendor (I have heard nothing bad about Keyboard Studio and have had good dealings with them myself--ed).

BAD GUYS--MICRO ARCHITECT. These guys were written up as bad before, and confirmed several times since. Their programs will not work properly. There are many bugs, parts of programs don't work, round off errors, etc. This junk is made for the TRaSh, let's keep it that way. Stay away from them!

GOOD GUYS--Bob Gregory/ 12600 S. 73rd Av/ Palos Heights, IL 60463. One of the frustrations Ralph Munroe has had with HEC is that they can't, don't or won't differentiate between the needs of a "game player" verses a commercial user in their support. Ralph is doing productive data processing for his employer's in-plant print shop, requiring almost daily use of his system. The two times his system went down they occurred at the end of month processing period.

The first time one HEC had no computer tech; the other had one but he would not talk to Ralph until after 4PM at which time he would be willing to send it to Benton Harbor for repair. He wouldn't fix it on-site. Ralph called BH explaining the situation and shipped the board the next day. **FRANK O'NEIL** and his staff (more **good guys**) had it back in 8 days. Frank has also bailed Ralph out in another situation. Frank is the service manager of Heath Company.

Getting back to the story, the second time the system crashed, still with no local support, it was shipped to BH again. Ralph appealed to H-SCOOP for a back-up system in his area and received no response (thanks guys). He called me and I did a zip code search and dug out some names of individuals nearby, Bob Gregory being one with a similar system. Upon contacting Bob, he graciously surrendered his controller board until Heath returned Ralph's so that the productive work could be continued. Thank you Bob. A friend in need is a friend indeed. I felt this was worth printing to show there are a few very dedicated and good guys left to help out in very troubled times. Keep this in mind if someone needs help, as that someone may be you some day.

GOOD GUYS--Larry Reeve of Polybytes. I have received several reports on Larry, this one by Don

Simpson sums it up best. "I bought the 2.8 version (of Lucidata Pascal from Larry-ed) on your recommendation and was quite pleased with it except for the absence of the trancendental math functions. I called Larry Reeve as soon as I got his flyer on the update and he shipped it COD. I had it within five days. Larry has been extremely helpful in answering my questions on the phone at night when I am sure he had more important things to do. .." I have had similar experience with Larry and he is always willing to help.

GOOD GUYS BRIEFS--Dean Gibson of Ultimeth for all the work and support he has done and is doing with HDOS and his Monitor ROMs.
John Avritt, Manager of the Kenner, LA Heathkit store. His support of the Heath users, his ingenuity in finding and supplying excellent hardware, and his making available the store for computer related events and meetings all add to his nomination of a Good Guy.
Don Long of Environmental Control Systems, for support of the products he sells. Don has been keeping customers up to date with latest software and application notes on his PC12 and PC89 clock/ parallel boards for the H8 and H89 computers.
T & E ASSOCIATES for excellent support of the software they write and distribute.
SOFTWARE TOOLWORKS. From John Croft "Have they ever turned out a bad program or incomprehensible instructions?" Excellent support of the products they carry, and excellent products available.
Brian Hawkins, manager of Vancouver WA Heathkit Center for his extraordinary after sales support. This type of support is what is needed, and will cause repeat business from a customer.
Burt Hulland for his MX-80 device drivers. They are excellent and his support is great!

BAD GUYS--LIFEBOAT ASSOCIATION--AGAIN!! I think of all the bad guys, Lifeboat tops the list for number of mentions in this newsletter. This writeup from Robert Williamson is so good, I will print it verbatim. "Lifeboat Assoc. has got to be one of the most incompetent marketing and/or customer service operations I have ever dealt with. It took two months and $14.00 worth of long distance phone calls, since they have no WATTS line and frequently forget that you are on hold, just to get them to send me the documentation for an update to a program. They were paid in advance for the upgrade. They sent the disc but only sent half the doc. This was obvious because the new function that prompted me to get the upgrade was the one for which the doc was missing.

"You may mention to your followers that Lifeboat Associates idea of "full service" means that if you subscribe to Lifelines, their magazine and manage to find out about a new version of a program (that you just paid $300 for) you may entice them to send an update for the paltry sum of $35 paid in advance. Then if you are persistant you can also get the documentation that goes with the upgrade. Heaven help the faint at heart for they shall receive headaches as their rewards. This is unfortunate since they are the only source for several very good programs."

BAD GUYS--Plantation Florida Heathkit store manager. This again, from Robert Williamson. "The manager of this place seems to be a pathological liar. On several occasions he has lied to me and even to the whole Heathkit User's Group (BUG). He can always come up with some excuse for not having any stock on spare parts or a technician to fix things, but that's so late now that several of us have bought machines for our business based on his promise of same day service and/or replacement.. He doesn't even stock small parts like connectors or voltage regulators since he doesn't get enough turn over on his stock to make it worthwhile to him. That's what he told me the reason was, but I've sold over $50 in connectors to the club alone, because he refused to stock them. These were replacements for Heath cables that went bad and since he didn't have spares OR parts we were stuck with $3,000 or more worth of useless computer.

"I thought it was going to be a temporary thing with him but it's his normal way of doing business... At least two of us have had to buy second computers because it was faster than getting ours repaired and we need them for our business. I am getting around this problem by selling my business system and going to an S-100 system, as are two other guys in the club. We lost several members because of this problem. They sold their H89's and bought other computers."

If this is the way you guys feel, write and call Heath. My local Heath store had very similar problems, only people started doing something about it. Suffice it to say, **that** manager is no longer there, and I hope no longer at ANY Heath position.

CLASSIFIEDS

FOR SALE--Anderson/ Jacobson AJ-841 printer. Like a Selectric II, except RS-232 Serial at 15 CPS. $800 with tractor feed and manuals. Like new. Ray Albrektson/ 619 W. Dover/ San Bernardino, CA 92407.

FOR SALE--H-14 printer. $275 of best offer (plus shipping). Robert Koestler/ 640 Trephanny Ln/ Wayne, PA 19087. (215) 337-6763 days or 687-3194 nights.

FOR SALE-- H77 with 2 40 TK Siemens drives. Excellent condition $625 or best offer. Heath 16K add on memory for H89, $75. Bob Flocchini/ 1810 Myrtle Pl/ Davis, CA 95616/ (916) 752-7090 days, 756-3296 nights.

FOR SALE--ADDS Regent 100 CRT Terminal. Like new, excellent condition. Best offer. Tom Roche/ 7 Winkel Way/ Ballstonlake, NY 12019/ (518) 877-5189.

WANTED-- QUME Sprint Micro 3 or "S3" printer, either complete unit or any parts. Will buy or swap. Call Collect, Dusan/ (713) 729-5500 after 8PM. 5416 W Bellfort/ Houston, TX 77035.

FOR SALE--1 Wangco disk drive. Mechanically perfect but electronically fried due to power reversal. Make offer? Don Long/ 9319 Willowview Lane/ Houston, TX 77080/ (713) 464-1717.

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PHONE HOURS: 9AM to 9PM as I am available, except sunset Friday to sunset Saturday (Seventh Day Adventist). NO COLLECT CALLS.